

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Developed and commissioned automation control systems for bespoke manufacturing machinery, achieving a **15% reduction** in assembly cycle time through integration of **Rockwell Automation PLCs** and **Ignition SCADA**.
- Automated pharmaceutical production workflows by developing **VBA scripts** for process parameter adjustment and **OSI PI Historian** data logging, ensuring **99.95%** compliance with regulatory audit trails.
- Designed and implemented a **Cisco-based networking infrastructure** for distributed automation controllers, improving inter-device communication latency by **20%** and enabling secure remote access.
- Enhanced system reliability for autonomous aqua drone navigation software by implementing **Python-based monitoring tools** and **Splunk dashboards**, reducing critical incident response time by **30%**.
- Supported and troubleshooted enterprise business systems, including **ERP integrations** and operational platforms, managing system configuration updates and coordinating **User Acceptance Testing (UAT)** across cross-functional teams.
- Managed change control processes for production systems, documenting technical requirements and post-implementation updates, which contributed to a **98% successful deployment** rate for new automation features.

SKILLS

Automation Platforms – Rockwell, Ignition, Siemens, OSI PI Historian, VBA

Control Systems – PLC Programming, SCADA Systems, Robotics

Software Development – Java Spring Boot, Python, REST APIs, .NET

Infrastructure & Networking – Cisco Networking, VMware, AWS, Docker, Linux

Data & Monitoring – SQL (PostgreSQL, MySQL, Oracle), Splunk, CloudWatch, Dynatrace

Project Management – Agile/Scrum, Jira, Confluence, Change Control, UAT

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Autonomous Aqua Drone Navigation System (HCL Technologies)

- Engineered a Python-based control module for autonomous aqua drone fleet management, integrating real-time sensor data and **AWS Lambda** for dynamic route optimization.
- Developed **REST APIs** using Java Spring Boot to enable seamless communication between ground control stations and the drone fleet, processing over 10,000 status updates per hour.

- Implemented comprehensive monitoring and alerting using **Splunk and CloudWatch**, achieving a **99.9% uptime** and reducing system downtime by **25%** through proactive incident detection.

Pharma Production Automation & Compliance Suite

- Developed custom **VBA macros** and integrated with **OSI PI Historian** to automate critical parameter adjustments in pharmaceutical manufacturing lines, ensuring data integrity and reducing manual error rates by **40%**.
- Designed and implemented a **SQL database schema** (PostgreSQL) for storing and retrieving historical production data, facilitating compliance reporting and **GDPR-aware** audit trails.
- Collaborated with business analysts to define operational requirements and supported UAT for the automation suite, leading to a **10% improvement** in overall production throughput.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)